Landon Bakken

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Education

University of Wisconsin Madison **Sept 2024 – Present**
Bachelor in Computer Engineering, expected July 2028 *Madison, Wisconsin*

Relevant Coursework: ECE 252, 352, 210, 230, 354 — CS 240, 300, 400

Experience

Wisconsin Racing Formula SAE **Sept 2024 – Present**
Controls Lead *Madison, Wisconsin*

2026 Electric Car

- Created a software, driver, and physics in loop simulation that contributed to **1st place in Overall Design**, reducing on-track tuning/development time by **64.4%**, and allowed for **80%** more control systems developed
- Developed a gradient descent based tuning system that can tune more parameters faster and more accurately
- Developed an unconventional traction control system that resulted in **3rd place Autocross** and **5th place Efficiency** due to both improved lap times and reduced energy consumption
- Developed launch control improved average acceleration by **8.5%**, resulting in **1st place Acceleration**
- Made a speed estimation system that enabled all control systems to function even when all wheels are slipping
- Created a dependable wireless telemetry system that drove endurance strategy

2025 Electric Car

- Developed and tuned traction control with load transfer feedforward and slip error feedback controller - reduced laptimes resulted in **2nd place Autocross**
- Developed and tuned torque vectoring based on speed and steering angle that was well integrated with traction control. Recognized by the **Multimatic Vehicle Dynamics Award**
- Developed driver interface relating charge and distance left in endurance that was specifically recognized for **1st place in Overall Design**
- Managed a multi-bus CAN communication system

Combustion Car

- Developed systems to tune engine cylinder air to fuel ratio, resulting in a **13.2%** power increase, and **3rd place in Efficiency**
- Made torque model that allows for precise and immediate torque cuts for traction control, faster pneumatic shifting, and faster throttle response, contributing to **3rd place in Overall Design**

Mercury Marine **May 2026 – Present**
Embedded Controls Engineer *Oshkosh, Wisconsin*

- Removed sensor noise from booster pump overshoot, reduced peak-to-peak noise by **45%**
- Developed a navigation program with demo features to reduce design cycle time by **1 year**
- Hull simulator visualizer to accelerate development of a virtual anti-collide system

Personal Projects

Machine Learning | *Neural Networks, Gradient Descent, Python* **Dec 2024 – Jan 2025**

- Built a machine learning model and training system from scratch using gradient descent in Python

Multiplayer Networking | *Networking, C#, Unity, Documentation* **Nov 2023 – Dec 2024**

- Developed a low-latency, peer-to-peer multiplayer system for Unity using UDP, TCP, and HTTP protocols, with comprehensive documentation for users without networking experience

Technical Skills

Computer Languages: Simulink, Python, C#, Java, C++, Javascript, Verilog, CSS/HTML
Manufacturing: SLA and FDM 3D printing, Laser Cutting, Soldering, CNC
Development Tools: Git, VS Code, Kvaser, Unity, SVN
3D Design: Blender, Fusion, Solidworks
Concepts: PID Controllers, CAN Protocol, Traction Systems, Signal Filtering, Networking, Vehicle Dynamics, Machine Learning